

Impact of bioinformatic pipeline selection on microbial community inference: a comparative analysis across pipelines including MetagenApp

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TAKE-HOME MESSAGE

Biological signal dominates pipeline effect

88.9%
SAMPLE IDENTITY

vs.

1.90%
PIPELINE CHOICE

High inter-pipeline concordance · Mantel $r = 0.94$

INTRODUCTION

Bioinformatic pipeline choice can materially alter the microbial community inferred from the same underlying sequencing data.

Prior cross-pipeline comparisons have often used mismatched reference databases or preprocessing, confounding true pipeline effects with implementation artifacts.

We ask what changes when MetagenApp, mothur, and QIIME 2 are benchmarked head-to-head under identical reference, filtering, and primer-trimming conditions.

METHODS

- Dataset: 36 paired-end MiSeq 16S V3–V4 samples (pharyngeal & nasal), 1.52M raw reads (González-García et al. 2024).
- Primer trimming: cutadapt, anchored 341F/805R — identical across all three pipelines.
- Reference: common curated SILVA 138.2 NR99 for all three (matched length, quality, degenerate-base filters).
- Feature inference: MetagenApp & mothur use 97%-similarity clustering (VSEARCH); QIIME 2 infers ASVs (DADA2).
- Compute: each pipeline run independently, same hardware, no concurrent processes.
- Diversity: alpha rarefied to 1,414 reads/sample; beta via Bray-Curtis PCoA/PERMANOVA (paired Friedman–Wilcoxon, Holm-corrected).

FIGURE 1 · WORKFLOW OVERVIEW

MetagenApp Pipeline: Paired-End FASTQ to Analysis

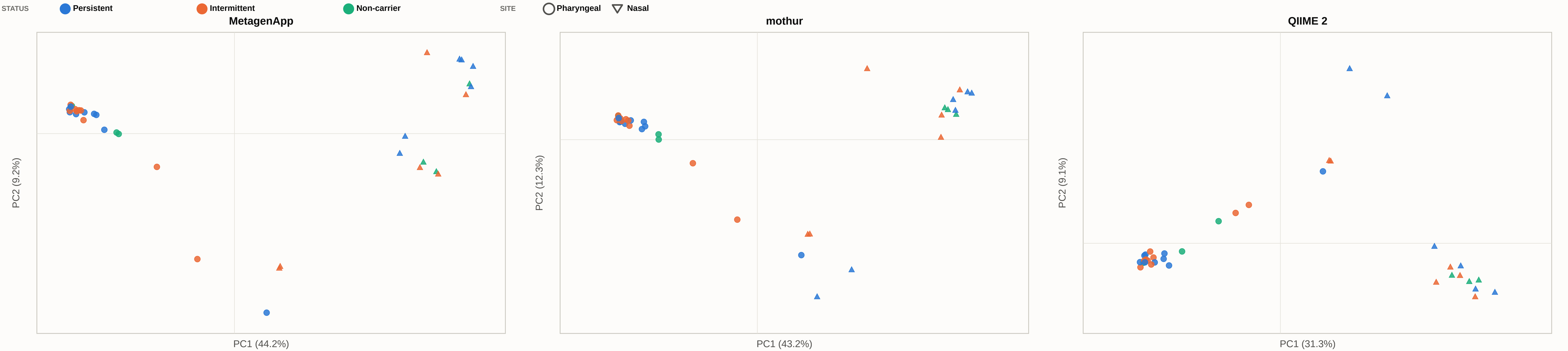


RESULTS

FIGURE 2 · BETA DIVERSITY

Community structure remains highly concordant across pipelines

Bray–Curtis PCoA by carrier status and sampling site



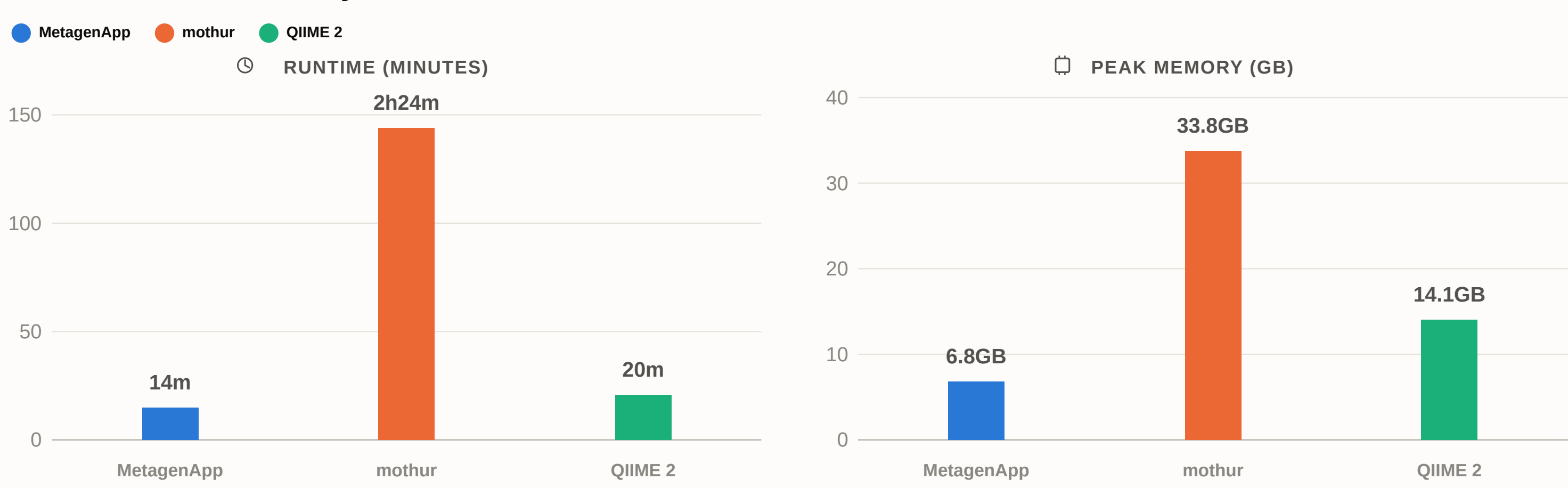
$r = 0.94$

INTER-PIPELINE CONCORDANCE (MANTEL)

Bray–Curtis PCoA · relative abundance · $n = 36$ matched samples per pipeline

FIGURE 3 · COMPUTATIONAL COST

Runtime & Peak Memory



METAGENAPP VS. MOTHUR

9.7×

FASTER (RUNTIME)

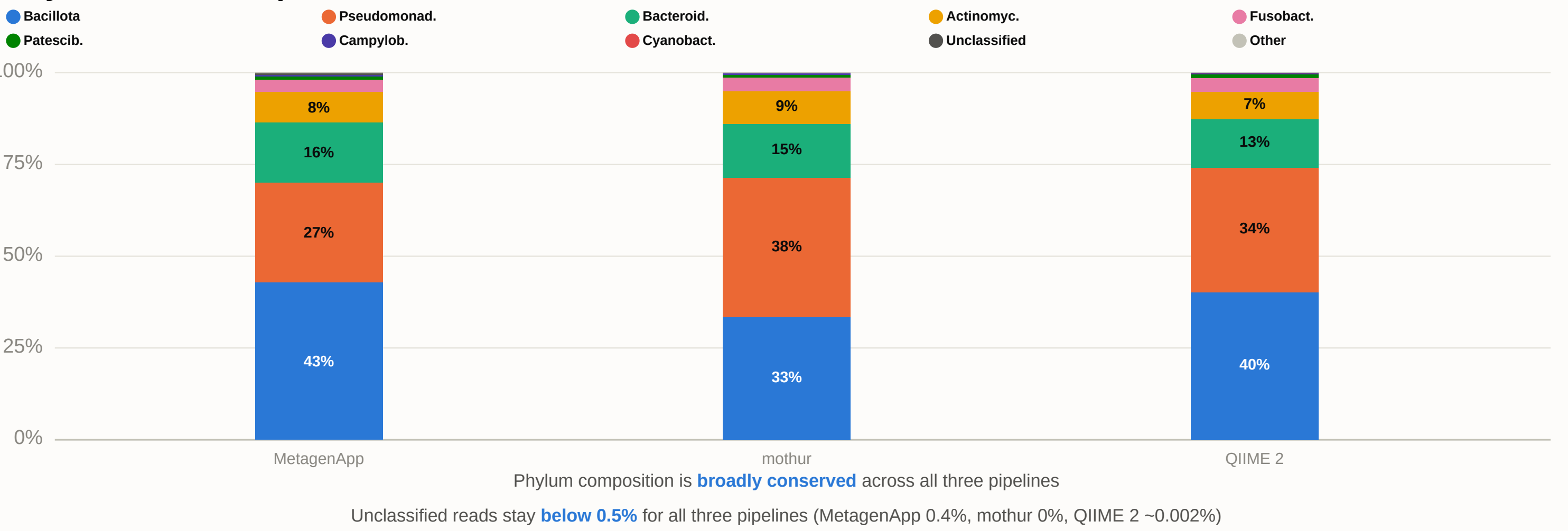
~5×

LESS PEAK MEMORY

Each pipeline run in isolation, same hardware · SILVA 138.2, 36 samples

FIGURE 6 · TAXONOMIC COMPOSITION

Phylum-Level Composition



DISCUSSION & CONCLUSIONS

BIOLOGICAL SIGNAL IS ROBUST

Sample identity explains substantially more variation than pipeline choice.

METAGENAPP REDUCES COMPUTATIONAL COST

Comparable ecological patterns recovered with substantially lower runtime and memory use.

PIPELINE EFFECTS ARE METRIC-DEPENDENT

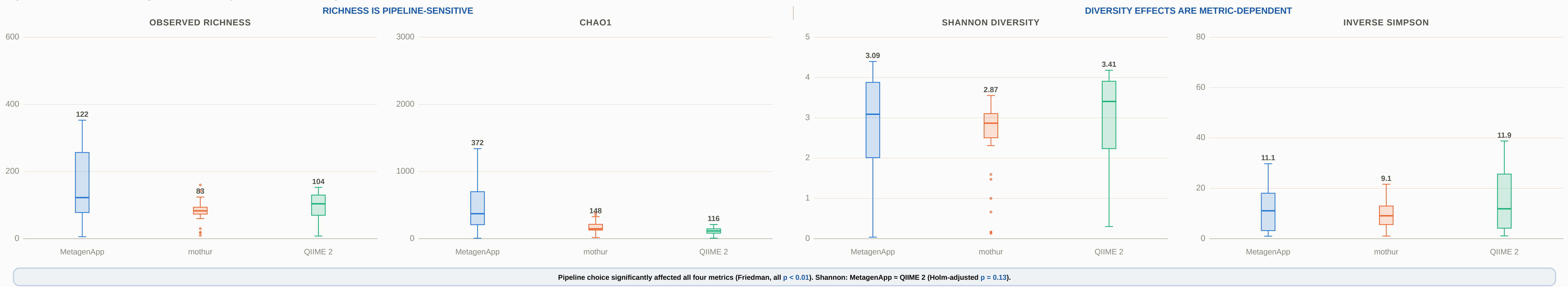
Richness estimates are particularly sensitive; Shannon converged between MetagenApp and QIIME 2.

NEXT STEP: EVALUATE ASV-STYLE INFERENCE

To test whether richness inflation can drop while preserving MetagenApp's efficiency.

FIGURE 4 · ALPHA DIVERSITY

Pipeline effects on alpha diversity are metric-dependent



Pipeline choice significantly affected all four metrics (Friedman, all $p < 0.01$; Shannon: MetagenApp > QIIME 2 (Holm-adjusted $p = 0.13$).

Rarefied to 1,414 reads/sample · $n = 36$ per pipeline · box = median & IQR, whiskers = 1.5×IQR · values above boxes are medians

TABLE · BENCHMARK AT A GLANCE

	MetagenApp	mothur	QIIME 2
Final feature count	12,599 features	7,552 OTUs	1,332 ASVs
Reads retained (final table + raw reads)	69.9%	68.2%	66.5%

TRY METAGENAPP

Code · Documentation · Reproducible workflow

github.com/mijailcampos/metagenapp



REFERENCES

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